

Active Normal Faces and Batched Geometric Descent

Certified improvements for unknot recognition

Research continuation for Vladimir Reshetnikov

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Scope. Exact support certificates, a structural obstruction to root-level propagation, simultaneous Pachner reductions, and dynamic vertex-link scoring. The complete recognizer's general quasipolynomial complexity remains unproved.

Reproducible baseline.

66098968e88bba797143ac1bf7ad0ac4c5f697df

<https://github.com/VladimirReshetnikov/ProveIt>

Abstract

We develop and implement certified improvements at two interfaces of the ProveIt unknot-recognition programme. The principal geometric result is an exact polynomial reduction of optimal vertex-link-peeled Euler scoring, over every subbatch of a fixed tetrahedron-disjoint family of coherent 3–2 Pachner moves, to maximum-weight matching. A minimum-omission optimum omits at most one move per relevant interior vertex. The maintained pulling construction has exactly two interior vertices, so its optimum omits at most two moves and its matching stage has a linear arithmetic-time selector. The conflict graph of all distinct legal central edges has degree at most nine; consequently a packed optimum retains at least $\max\{0, \lceil M/10 \rceil - 2\}$ of M current candidates. We also prove relative-boundary commutation, implement one simultaneous certificate, and maintain exact peeling minima with deterministic balanced trees. A genuine 32-tetrahedron solid-torus example shows why adding singleton peeled scores would be incorrect.

On the linear-programming side, a primal–dual pair certifies the entire active support of a rational nonnegative matching cone. A homogeneous lift produces that pair or an ordinary small negative certificate. An adaptive quadrilateral search has at most $2^{\rho+1} - 1$ nodes in terms of an intrinsic active ambiguity rank ρ . However, explicit local normal-coordinate directions prove $\rho \geq t - v$ at unrestricted anchored roots, and $\rho \geq 16 \max\{1, n\} - 2$ on the maintained n -crossing pulling source. This is a structural obstruction to obtaining a universal polylogarithmic bound from active support alone.

The additive implementation includes independent consumers, exact replay fixtures, paired timing data, and a pinned integration patch. Four 64-move workloads show approximately 117–123-fold producer improvements over literal sequential replay, while the enlarged support LP regresses against small unary probes. These are subroutine results. We identify the remaining witness-coverage, transition-count, anchor, and bit-complexity theorems required for a complete quasipolynomial recognizer, and propose a concrete research programme to address them.

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1 Research question and precise contribution

The computational objective is a complete unknot recognizer with a quasipolynomial worst-case bound in the crossing number. The maintained ProveIt implementation combines exact filters, group-theoretic methods, normal-surface certificates, and a complete fallback. Many local operations are now substantially faster than their first implementations. The central research issue is whether the number and encoded size of the states visited can be bounded without expanding exponentially large surfaces or returning to an exhaustive search over all normal-coordinate sectors.

This continuation works on two interfaces where the existing research packages leave concrete opportunities. In the normal-coordinate search, syntactically allowed quadrilateral types need not be geometrically possible. A certificate of the entire active support makes it possible to remove these spurious choices in one operation and define the remaining ambiguity intrinsically. In geometric descent, many currently legal Pachner reductions have disjoint tetrahedral supports. They can be verified and applied together, avoiding repeated emission and validation of almost identical triangulations. Vertex-link peeling then requires careful collective scoring: disjoint tetrahedral supports can still share a global vertex, so their peeled scores need not add.

1.1 What is established here

The mathematical results fall into five groups.

1. A primal–dual description of the maximal active support of a nonnegative homogeneous matching cone. A single aggregate dual witness certifies every universally zero coordinate. Combining this with a positive objective gives a complete support certificate, independently checkable without rerunning the search procedure. The underlying polyhedral principle is classical strict complementarity; the contribution is its concrete certificate use at this repository interface.
2. An ambiguity parameter for restricted quadrilateral search, together with a finite search-tree bound, and a negative result: on an unrestricted one-vertex anchored root that has a positive-Euler witness, the ambiguity rank is at least $t - 1$. More generally it is at least $t - v(T)$, and at least $16 \max\{1, n\} - 2$ on the maintained pulling source. The obstruction is explicit and comes from local directions invisible to all face-arc equations.
3. A simultaneous relative-boundary replacement theorem for disjoint 3–2 regions. The conflict graph of distinct legal degree-three central edges has maximum degree at most nine. A greedy maximal batch therefore contains at least a tenth of the current candidates. The cocycle transport and unpeeled cell changes commute and add.
4. A deterministic data structure that maintains vertex-link minima under repeated local replacements. Cached thirteen-record prefixes give constant-operation hypothetical singleton scoring, while balanced-tree updates replenish these prefixes after commits. Collective scoring accounts for shared vertices. A realized solid-torus example proves that singleton peeled gains cannot simply be added.
5. An exact maximum-weight-matching reduction for the best peeled-Euler subbatch of any fixed disjoint family. A minimum-omission optimum omits at most one move per relevant interior vertex. On the maintained pulling source, which has exactly two interior poles, a linear arithmetic-time selector omits at most two moves. Thus the packed optimum retains at least $\max\{0, \lceil M/10 \rceil - 2\}$ current candidates.

All statements are scoped to their actual inputs. A positive-Euler relaxed matching vector may violate the quadrilateral constraints. A legal Pachner move preserves the ambient manifold, but coherent surfaces transported through it can change topology. A positive total Euler characteristic is therefore not an unknot certificate. The supplied code retains the repository’s independently verified component-disc query as the relevant terminal geometric test.

1.2 Position relative to the latest incoming work

The source revision inspected is 66098968e88bba797143ac1bf7ad0ac4c5f697df. Its `fast/` and `synthesis/` files used here were verified against their Git blob identities. The three relevant archives in `docs/incoming/` were also verified byte-for-byte against the repository’s recorded blobs: the dual-certificates report, the support-sensitive-certificates report, and the geometric-transport report, all dated 9 October 2026 [2, 3, 4]. They are prior work, not new deliverables of this continuation.

The dual report introduced independently replayable LP objective certificates and forced-type propagation. The support report sharpened component-incidence bounds and primitive ray certification. The transport report established the exact five-height Pachner score, coherent cocycle transport, and the thirteen-record lemma for a *fixed* scoring pass. It explicitly left that last lemma unimplemented and warned that a short prefix must be replenished after repeated deletions. The present dynamic construction supplies that missing maintenance mechanism and connects it to simultaneous geometric replacements.

The package contains a pinned supporting snapshot, a small overlay for the new modules, an integration patch, exact proof fixtures, retained measurements, and reproduction scripts. It is prepared for review and integration; no remote repository modification is part of this task.

1.3 The literature does not remove the global gap

Lackenby’s technical announcement describes a hierarchy strategy with a quasipolynomial running-time target [12]. The particular Oxford handout gives $2^{O((\log n)^3)}$, which is $n^{O((\log n)^2)}$. This should not be silently replaced by the sharper $n^{O(\log n)}$ expression sometimes quoted in public descriptions. Both are quasipolynomial, but the exponent matters in a rigorous cost ledger.

The July 2026 preprint gives a substantial hierarchy-based recognition algorithm and new certificate results [13]. Its discussion of running time explicitly leaves algorithmic precision and speed-up for further work. Proposition 10.2 bounds the length of a partial hierarchy by a linear function of quadrilateral handle complexity; it does not establish the missing polylogarithmic effective-depth statement for the code developed here.

The normal-coordinate part also has important established precedents. Burton–Ozlen’s tree traversal uses LP feasibility and greedy support minimization to obtain a connected positive-Euler surface [8]. Burton’s admissible-face theory identifies compatibility with the geometry of nonnegative normal cones [9]. The new support certificate must be understood against those results. We neither claim that LP support reduction is new nor infer a polynomial recognition algorithm from the polynomial solvability of a single LP.

2 Geometric, algebraic, and encoding conventions

2.1 Finite triangulations and local identifications

Let T be a finite generalized triangulation with t tetrahedra. Each tetrahedron has four numbered vertices. A face is either unpaired or is paired to another face by a permutation of the four local

vertices. The reciprocal pairing and its inverse are both recorded. Local corners, edges, and faces are distinguished from global equivalence classes under these identifications. In particular, several corners of a tetrahedron can represent the same global vertex.

The implemented geometric validator requires a connected orientable compact three-manifold with one torus boundary component. It checks reciprocal pairings, orientation consistency, edge identifications, vertex links, and the boundary's connectedness and Euler characteristic. An interior vertex link is a sphere; a boundary vertex link is a disc. These are the hypotheses under which the following normal-coordinate and Pachner operations are used. Establishing that an arbitrary supplied triangulation is the exterior of the user's knot is a separate provenance obligation, supported by the repository's diagram-exterior routines.

2.2 Normal coordinates

Write a standard normal vector as

$$x = (a_{iv}, q_{ij}) \in \mathbb{R}_{\geq 0}^{7t}, \quad 0 \leq i < t, \quad 0 \leq v < 4, \quad 0 \leq j < 3.$$

Here a_{iv} counts triangles cutting off vertex v in tetrahedron i , and q_{ij} counts quadrilaterals of type j . Across each paired face, the three normal-arc counts agree. These conditions form a sparse integer linear system $Ax = 0$. Each arc equation is a difference of two triangle coordinates and two quadrilateral coordinates, with coincidences combined. The additional quadrilateral condition requires at most one positive q_{ij} in each tetrahedron. It is this nonconvex condition that necessitates sector choices or a different global method.

On the linear matching space the Euler characteristic is a rational linear functional $c^T x$. One may calculate it by assigning each global edge-intersection count to a representative local edge and counting one copy of each glued normal arc. For integral admissible vectors the result is the integer Euler characteristic of the corresponding surface. The linear formula remains algebraically defined on relaxed vectors with incompatible quadrilaterals; those vectors do not necessarily represent embedded surfaces.

The full integral vector can be very large in value but small in encoding. Let B bound the bit length of its coordinates or local height differences. Algorithms here manipulate $O(t)$ binary integers, not one object per normal disc. Any theoretical polynomial LP bound refers to the full rational input bit length. The shipped exact simplex is an implementation choice with explicit resource limits, and has no polynomial pivot bound.

2.3 Anchors and vertex links

For a global vertex v , let C_v be its set of local corner records. The normal vertex-link vector L_v has one triangle at every corner in C_v and zero quadrilaterals. The maximal removable multiplicity is

$$m_v(x) = \min_{c \in C_v} a_c, \quad \text{peel}(x) = x - \sum_v m_v(x) L_v.$$

The peeled vector is nonnegative, obeys the same matching equations, and is admissible when x is. Choosing one triangle coordinate at each global vertex and setting it to zero gives an *anchor face*. Every peeled vector lies on some anchor face. A one-vertex triangulation has at most $4t$ raw corner anchors, with additional contractions possible as in the incoming dual report.

Peeling is distinct from division by the common divisor of quadrilateral coordinates. The latter can be useful for component-count queries, but it changes all coordinates and is not part of the

local-update theorem. For $\lambda_v = \chi(L_v) \in \{1, 2\}$ and $d_v = |C_v|$,

$$\chi(\text{peel}(x)) = \chi(x) - \sum_v \lambda_v m_v(x), \quad P(\text{peel}(x)) = P(x) - \sum_v d_v m_v(x), \quad (1)$$

where P is the number of normal discs before gluing. These identities also explain why vertices shared by disjoint local replacements can couple their peeled scores.

2.4 Integral cocycles and coherent fibres

An integral cocycle can be represented in each tetrahedron by four integer heights $h_{i0}, \dots, h_{i3}$. The signed height differences agree on globally identified edges. Adding the same constant to all four heights of one tetrahedron does not change those differences. Sorting its four heights as $h_a \leq h_b \leq h_c \leq h_d$ gives the coherent normal vector with $h_b - h_a$ triangles at a , $h_d - h_c$ triangles at d , and $h_c - h_b$ quadrilaterals separating $\{a, b\}$ from $\{c, d\}$. Equal heights cause zero counts and require no special perturbation.

The construction is the normal level set of the local affine height functions, with unit-separated levels taken modulo integral shifts. It is cooriented and its boundary and normal matching data agree under face gluings. A transport operation preserves the integral cohomology class but can split, join, or change components of this fibre. The incoming transport package contains a verified example in which one disc becomes a disc and two spheres [4]. Thus component analysis remains necessary after any geometric search policy.

2.5 Cost model and certificate scope

The comparison model counts arithmetic on integers separately from its bit cost. An operation on a bounded number of B -bit heights takes polynomial bit time in B ; multiplication by a corner count introduces $O(\log t)$ additional bits. We write $M(b)$ for a chosen integer multiplication cost and avoid treating a large integer as a machine word. Combinatorial indices use $O(\log(t + 2))$ bits.

Three kinds of output are deliberately distinct: an algebraic certificate about a supplied matching system; a geometric certificate relating two triangulations and transported cocycles; and a knot verdict tied to the original diagram. The first two can accelerate the third but do not replace its end-to-end proof requirements.

3 Active normal cones: complete support proofs and a structural obstruction

The previous continuation replaces some normal-ray enumeration queries by exact positive-Euler linear programmes and certifies their negative answers by Farkas multipliers. It also identifies a difficulty with a strong mandatory-type backdoor: a tetrahedron unused by a positive surface can retain several allowed types, even when that surface is easy to find [2]. An immediate response is to distinguish an allowed coordinate from one that is actually positive somewhere in the relaxed cone. This section makes that distinction exact, implements its certificate, and then proves a limitation of the resulting structural parameter.

The linear-programming ingredients are classical. Strict complementarity and maximal support belong to the Goldman–Tucker framework [5]; coordinate-face traversal and normal-surface LP search have substantial precedents [9, 8]. Our purpose is to give a source-sized proof interface, an explicit homogeneous construction compatible with the supplied exact solver, and a precise complexity bound for the adaptive tree. Familiar local matching identities then yield a linear

lower bound for its root parameter on genuine normal-coordinate inputs, including the maintained diagram-exterior construction.

3.1 One primal–dual pair identifies every active coordinate

Let $A \in \mathbb{Q}^{m \times N}$ and $c \in \mathbb{Q}^N$. Put

$$C = \{x \in \mathbb{R}^N : x \geq 0, Ax = 0\}, \quad S(C) = \{j : \text{some } z \in C \text{ has } z_j > 0\}. \quad (2)$$

The set $S(C)$ is the *active support of the cone*; it need not equal the support of a particular basic solution. Write $P_c = \{x \in C : c^T x > 0\}$. If P_c is nonempty, its active support is also $S(C)$. Indeed, if $p \in P_c$ and $z \in C$ has $z_j > 0$, then $p + \varepsilon z \in P_c$ for sufficiently small positive ε . The objective of z may be negative. Activity on the positive-objective part therefore cannot be inferred from the sign of an individual direction's objective.

Lemma 3.1 (Complete active-support alternative). *Exactly one of the following alternatives holds:*

1. *There are rational vectors $x \in \mathbb{Q}^N$ and $y \in \mathbb{Q}^m$ satisfying*

$$Ax = 0, \quad x \geq 0, \quad c^T x \geq 1, \quad s := A^T y \geq 0, \quad x + s \geq \mathbf{1}. \quad (3)$$

2. *There is a rational vector $h \in \mathbb{Q}^m$ such that*

$$A^T h \geq c. \quad (4)$$

In the first alternative, $x_j \geq 1$ exactly when $j \in S(C)$, and $s_j \geq 1$ exactly when $j \notin S(C)$.

Proof. For a pair satisfying (3),

$$x^T s = x^T A^T y = (Ax)^T y = 0.$$

Both vectors are nonnegative. Hence $x_j s_j = 0$ in every coordinate, while $x_j + s_j \geq 1$ ensures that exactly one of the two entries is at least one. For any $z \in C$, the same calculation gives $z^T s = 0$. Thus $s_j > 0$ forces $z_j = 0$ throughout C , whereas $x_j > 0$ explicitly demonstrates activity. This proves the support assertion.

Suppose $P_c \neq \emptyset$. For each $j \in S(C)$ choose a rational $z^{(j)} \in C$ with $z_j^{(j)} > 0$. Rational witnesses exist because the relevant linear systems are rational. Their sum z is positive on all of $S(C)$. Starting from a rational $p \in P_c$, choose a sufficiently small positive rational ε and then rescale $p + \varepsilon z$ to obtain $x_j \geq 1$ on $S(C)$ and $c^T x \geq 1$.

For every inactive coordinate j , the functional $e_j^T z$ is nonpositive on C ; in fact it is identically zero. The homogeneous Farkas alternative gives a rational $y^{(j)}$ with $A^T y^{(j)} \geq e_j$. Let $y = \sum_{j \notin S(C)} y^{(j)}$, with the empty sum interpreted as zero. Its residual is nonnegative everywhere and at least one off $S(C)$. Orthogonality with the constructed x makes it zero on $S(C)$. Thus (3) holds.

If $P_c = \emptyset$, homogeneous Farkas gives (4). Conversely, pairing that inequality with any nonnegative kernel vector gives $c^T x \leq h^T Ax = 0$, which excludes the first alternative. $\square$

The proof has a useful verification consequence. The checker does not need one infeasibility test per inactive coordinate, and it does not reproduce a relative-interior algorithm. It checks two matrix products, signs, and the coordinatewise coverage inequality in (3). One multiplier exposes all inactive coordinates simultaneously; one positive vector proves that no remaining coordinate has been silently omitted.

The claim is about the relaxed cone. For example,

$$A = (1, -1), \quad c = (1, 1), \quad C = \{(a, a) : a \geq 0\}$$

has both coordinates active and positive objective away from zero. If the two coordinates are declared incompatible, it has no admissible positive point. A complete active-support vector can thus be a particularly incompatible relaxation vector.

3.2 A single homogeneous query with a small negative output

The inherited solver accepts positive-objective queries over a nonnegative homogeneous kernel. We can use that interface directly. Introduce nonnegative variables

$$(x, y^+, y^-, s, u, v, \tau),$$

where x, s, u have N entries, y^+, y^- have m entries, and v, τ are scalars. Impose

$$Ax = 0, \quad s - A^T(y^+ - y^-) = 0, \quad (5)$$

$$x + s - u - \tau \mathbf{1} = 0, \quad c^T x - v - \tau = 0. \quad (6)$$

There are $3N + 2m + 2$ nonnegative variables and $m + 2N + 1$ equations. The objective is τ . A point with $\tau > 0$, divided by τ , yields (3) with $y = y^+ - y^-$. Conversely, split any multiplier from that certificate into its positive and negative parts, set $\tau = 1$, and use the nonnegative surpluses for u and v . The lifted positive query is therefore equivalent to $P_c \neq \emptyset$.

It is also possible to discard the large lifted dual after a negative answer. The following projection is useful because the consumer can remain an ordinary homogeneous-Farkas checker on the original matrix.

Proposition 3.2 (Projection of a lifted negative proof). *Let $(\alpha, \beta, \gamma, \delta)$ be a negative Farkas multiplier for the lifted positive- τ query, grouped according to the four displayed sets of equations. Then $\delta < 0$, and $h = \alpha/(-\delta)$ satisfies $A^T h \geq c$.*

Proof. The lifted column inequalities are

$$\begin{aligned} A^T \alpha + \gamma + \delta c &\geq 0, & A\beta &= 0, \\ \beta + \gamma &\geq 0, & \gamma &\leq 0, \\ \delta &\leq 0, & -\mathbf{1}^T \gamma - \delta &\geq 1. \end{aligned}$$

The equality $A\beta = 0$ comes from the two oppositely signed columns for each free dual variable. If $\delta = 0$, then $\beta \geq -\gamma \geq 0$. Pairing the first inequality with β gives

$$0 \leq \beta^T \gamma \leq -\sum_j \gamma_j^2.$$

Thus $\gamma = 0$, which contradicts $-\mathbf{1}^T \gamma - \delta \geq 1$. Hence $\delta < 0$. Finally,

$$A^T \alpha \geq -\gamma - \delta c \geq -\delta c,$$

and division by $-\delta$ gives the required original-cone dual. $\square$

The enlarged system has polynomial encoding length in the original rational input. Standard rational-LP algorithms consequently decide it in polynomial bit complexity, with polynomial-bit primal or dual witnesses. The executable producer uses the exact Bland-simplex routine inherited from the previous continuation [2]. Its arithmetic is exact and its configured pivot allowance is explicit; this implementation is not asserted to have a polynomial worst-case pivot count. The distinction is material: one large degenerate LP can cost much more than several small support probes. The experiments section reports that tradeoff, including unfavorable outcomes.

3.3 An intrinsic rank for the remaining incompatibilities

Let $\mathcal{G}$ be a collection of coordinate conflict groups. In normal coordinates these are the three quadrilateral types in each tetrahedron. A vector is admissible when each group contains at most one positive coordinate. The following definition also permits overlapping groups, since the proof uses only their incompatibility disjunctions.

Definition 3.3 (Active ambiguity rank). For C as in (2), let $V = \text{span}(C)$ and put

$$J(C) = \bigcup_{\substack{G \in \mathcal{G} \\ |G \cap S(C)| \geq 2}} (G \cap S(C)).$$

Define

$$\rho(C) = \dim \text{span}\{e_j|_V : j \in J(C)\}. \quad (7)$$

This quantity depends on the cone and the conflict groups, not on the LP solver's particular basic point. Universally zero coordinates do not contribute. Neither do independent directions carried only by coordinates outside every active conflict.

It is an exact linear-algebraic quantity once the active-support certificate has been checked. If $S = S(C)$, then

$$V = \{z : Az = 0, z_j = 0 (j \notin S)\}, \quad \dim V = |S| - \text{rank } A_S. \quad (8)$$

Indeed, a vector positive on S has a relative neighborhood in $\ker A_S$ that remains nonnegative, so the cone spans that entire kernel. Writing E_J for the indicated coordinate rows restricted to S gives

$$\rho(C) = \text{rank} \begin{pmatrix} A_S \\ E_J \end{pmatrix} - \text{rank } A_S. \quad (9)$$

Rank computation is useful for interpretation and complexity accounting; the certificate checker does not need to trust a reported rank in order to accept a positive witness or a negative search tree.

Theorem 3.4 (Binary active-face search). *Suppose C contains a point of positive objective. At each nonterminal positive node, compute complete active support and choose any incompatible active pair i, j . Recurse on*

$$C \cap \{x_i = 0\} \quad \text{and} \quad C \cap \{x_j = 0\}.$$

Terminate at a nonpositive cone or at a positive cone whose active support is compatible. This decides admissible positive-objective existence. Its complete binary tree has at most $2^{\rho(C)+1} - 1$ nodes. With polynomial rational LP, its bit cost and certificate length are $2^{\rho(C)} \text{poly}(L)$, where L is the original encoded input length.

Proof. Every admissible vector satisfies $x_i = 0$ or $x_j = 0$, so the two children cover all admissible possibilities. Their overlap is harmless. A coordinate deleted using an active-support dual is zero on the entire current cone; such a deletion preserves every admissible candidate. A negative dual excludes positive objective. At a compatible positive leaf, the active primal vector is itself admissible. These observations prove correctness.

For the bound, let C' be a child whose positive-objective part is nonempty, and put $V' = \text{span}(C')$. Coordinate restriction cannot create new active coordinates, so $J(C') \subseteq J(C)$. Let

$$W = \text{span}\{e_k|_{V'} : k \in J(C')\}.$$

In the child $x_i = 0$, restriction from V to V' kills $e_i|_V$. That functional is nonzero because i was active, and it belongs to W because i was part of an active conflict. The restriction map on W therefore has nontrivial kernel. Its image has dimension at most $\rho(C) - 1$, and the defining functionals for $\rho(C')$ lie in that image. Consequently

$$\rho(C') \leq \rho(C) - 1.$$

The same argument applies to the other child.

If $\rho = 0$, there is no active incompatible pair: any active coordinate in such a pair would be a nonzero defining functional. Hence every positive branching node has rank at least one. Along a path, ranks decrease at each edge whose child remains positive. A final negative leaf can occur after the last rank-one branching node, so the full path still contains at most $\rho(C)$ branches. Binary tree counting yields the stated node bound. Every query and arithmetic witness has polynomial encoded size, and a negative tree retains both children at each split, giving the time and certificate estimates. $\square$

The proof makes no assumption that the selected coordinate-zero face drops dimension by exactly one. A child may lose several coordinates at once, which only improves the bound. It also permits additional early terminal rules. In particular, a small ordinary LP may first return a negative dual or an already admissible basic vector. Stopping in either case shortens the tree. The implementation uses this precheck by default and invokes the larger active-support LP only for an incompatible positive relaxation.

3.4 Normal anchors and the precise scope of a conclusion

For a finite triangulation, choose one zero triangle coordinate at every global vertex. Let m_v be the number of local triangle corners representing vertex v . The complete anchor product has size

$$\mathcal{A}(T) = \prod_v m_v, \quad \sum_v m_v = 4t.$$

In the one-vertex case, $\mathcal{A}(T) = 4t$. Restricting the standard matching matrix and an exact Euler covector to the surviving columns gives the anchored systems used above. Every canonical admissible normal vector belongs to at least one such anchored cone.

If $\rho_{\max}$ bounds the roots that require positive relaxed search, the complete anchored positive-Euler query therefore has the conditional cost

$$\mathcal{A}(T) 2^{\rho_{\max}} \text{poly}(t) \tag{10}$$

with a polynomial rational-LP backend. A negative proof contains every anchor tuple in the independently reconstructed product, with a complete binary proof tree for each tuple. Missing anchors, a missing child, or an uncertified coordinate deletion invalidate coverage.

A positive conclusion here is a canonical admissible normal vector with positive Euler characteristic. A maximal-support vector may have several components; clearing denominators and dividing by the gcd does not turn it into an extreme ray or prove connectedness. The maintained geometric component/disc endpoint is still required to identify an essential disc. Conversely, a negative positive-Euler query is one ingredient in a complete normal-surface recognition architecture, whose provenance and crushing transitions must also be justified [8, 2]. The new module does not convert either status directly into a knot verdict.

3.5 A linear obstruction already present in genuine root cones

The active parameter removes identically zero coordinates after restrictions. At an unrestricted normal root, however, there is a universal source of activity. Let $v(T)$ denote the *total* number of global vertices; this includes boundary vertices and is distinct from the number of interior vertices used in the batch-selection results. Fix one zero triangle anchor per global vertex, and let u be the number of tetrahedra containing at least one selected anchor.

Theorem 3.5 (Unrestricted anchored normal-cone obstruction). *Allow all three quadrilateral types in every tetrahedron and impose only the matching, nonnegativity, and chosen triangle-anchor equations. Then every quadrilateral coordinate is active, and*

$$\rho(C) \geq t - u \geq t - v(T). \quad (11)$$

This conclusion does not require positive Euler feasibility. If the positive-Euler part is nonempty, all quadrilateral coordinates are also active on that part. In particular, a one-vertex root satisfies $\rho(C) \geq t - 1$.

Proof. Let r have every triangle coordinate zero and every quadrilateral coordinate one. Each triangular face receives one arc of each normal arc type from these three quadrilaterals. Thus all face-matching equations hold. Every selected triangle anchor vanishes, so $r \in C$ and every quadrilateral coordinate is active. Each tetrahedron consequently belongs to an active three-type conflict.

For a tetrahedron a containing no selected anchor, define w_a to have value +1 in its four triangle coordinates, value -1 in its three quadrilateral coordinates, and zero elsewhere. A local normal arc count has the form $T + Q$. Its change under w_a is therefore $1 - 1 = 0$, on every face of a . This verifies the matching-kernel identity also when faces of the same tetrahedron are identified. Both r and $r + w_a$ are nonnegative and satisfy all chosen anchors. Hence $w_a \in \text{span}(C)$.

The quadrilateral image of w_a is -1 precisely in the three coordinates of tetrahedron a . These images have disjoint supports as a ranges over the $t - u$ unanchored tetrahedra, so they are linearly independent. All of their coordinate functionals occur in the definition of $\rho(C)$ because all quadrilateral types are active. Thus the map defining the ambiguity rank has rank at least $t - u$. There is at most one selected anchor per global vertex, and hence $u \leq v(T)$, proving the displayed inequality.

Finally, if $p \in C$ has positive Euler characteristic, then $p + \varepsilon r$ still has positive Euler characteristic for sufficiently small positive ε , and it has every quadrilateral coordinate positive. This proves the last activity assertion. $\square$

The vectors r and $r + w_a$ are incompatible relaxation vectors. The proof does not assert that they are embedded surfaces. Their role is to expose freedom that an LP relaxation necessarily retains before useful type restrictions or admissibility implications have been imposed.

The maintained source constructor gives a direct family-specific form of the obstruction.

Corollary 3.6 (The pulling diagram-exterior source). *Let a classical one-component diagram have n crossings and put $n' = \max\{1, n\}$. For the maintained pulling construction, with its checked compact knot-exterior interpretation, every unrestricted anchor root satisfies*

$$\rho(C) \geq 16n' - 2. \quad (12)$$

Proof. The source has $4n'$ truncated corner cells and uses five tetrahedra in each cell, so $t = 20n'$. Each cell contributes two truncation triangles to the boundary, giving $f_\partial = 8n'$. These counts

are part of the maintained pulling template and source correspondence [1]. The boundary is a triangulated torus; counting incidences and using its Euler characteristic gives

$$e_{\partial} = \frac{3}{2}f_{\partial} = 12n', \quad v_{\partial} = e_{\partial} - f_{\partial} = 4n'.$$

The pulling subdivision introduces no interior vertices beyond the two finite pole classes of the corner construction. Its remaining vertices are boundary cut points. Thus $v(T) = 4n' + 2$. Applying (11) yields $\rho(C) \geq 20n' - (4n' + 2) = 16n' - 2$. For $n = 0$, the constructor uses its specified one-curl representative of the crossing-free circle, so the same count with $n' = 1$ applies. $\square$

These lower bounds rule out a universal polylogarithmic bound for this *unrestricted-root* rank along the displayed source family. They do not give a lower bound for the recognizer's actual runtime: a negative ordinary LP or an early admissible point can finish a stage before the active query runs. They also do not rule out small parameters after mandatory propagation, certified type restrictions, or topology changes. The distinction is concrete. The earlier continuation proves that its one-vertex layered Fibonacci family requires no guessed types under the mandatory rule [2], whereas Theorem 3.5 gives a linearly growing unrestricted active rank on the same family.

3.6 Implementation contract and research consequence

The additive implementation has three levels. The rational-cone primitive returns a complete active-support pair, an ordinary negative multiplier, or an inconclusive resource-limit outcome. The conflict-group search adds complete binary proof trees. The normal wrapper reconstructs finite matching equations, the Euler covector, and the full anchor product, and binds its final certificate to the source triangulation. Its independent checker imports neither the producer nor an LP solver. Seventeen focused tests cover arithmetic alternatives, hidden active coordinates, malformed rationals, incomplete branches, cancellation, anchor/source substitution, and actual normal positive and negative controls. In particular, the positive meridian control is passed onward to the maintained geometric essential-disc counter.

The mathematical reduction and the executable backend have different roles. The active-support pair is a short, complete interface for deleting inactive coordinates. The binary theorem gives an honest intrinsic bound at any restricted cone. The root obstruction identifies an unavoidable linear source of ambiguity before such restrictions. Further progress must therefore explain how geometric information, admissibility implications, or a controlled transition removes that freedom, or why a verified endpoint is found before it matters. Merely replacing allowed masks by active supports cannot supply the missing global quasi-polynomial theorem.

4 Commuting Pachner reductions and one combined certificate

4.1 The exact formal-region contract

A legal 3–2 site is an interior edge whose three local occurrences belong to three distinct tetrahedra. Label its endpoints A, B and the three belt vertices C, D, E . The old tetrahedra have formal vertex sets

$$ABCD, \quad ABDE, \quad ABEC.$$

Their three internal faces have the consistent shared formal labels. The replacement consists of

$$ACDE, \quad BCDE,$$

glued along CDE . Both fillings are balls with the same six labelled boundary facets. The contract retains the complete permutation at every boundary gluing. It permits global identifications of formal boundary vertices, edges, and facets; it does not identify any extra local occurrence with the central edge.

The distinct-tetrahedron and formal-label conditions matter. An arbitrary degree-three entry in a poorly checked edge census is not enough. The producer reconstructs the local labels, and the independent consumer checks the old and new region interiors and all external attachment permutations. This is the same local topological contract used by the maintained individual-move checker.

Theorem 4.1 (Simultaneous relative-boundary replacement). *Let E be a collection of legal 3–2 regions in a finite triangulation T , with pairwise disjoint sets of tetrahedra. Replace every region in E by its two-tetrahedron filling and transfer its six boundary-facet gluings. Then:*

- (i) *the result is homeomorphic to T relative to the unchanged external boundary;*
- (ii) *every order of performing the individual replacements is legal and produces the same labelled quotient after the explicitly determined permutation of tetrahedron indices;*
- (iii) *a supplied integral cocycle has the same transported signed edge values and coherent normal coordinates in every such order;*
- (iv) *the raw Euler changes and raw normal-piece changes add over the regions.*

Regions may share global vertices, global boundary edges, or facets paired between different regions.

Proof. Before taking the quotient, partition the tetrahedra into unchanged tetrahedra and the formal bipyramids. A ball filling of a bipyramid comes with its boundary triangulation and vertex labels. The two fillings are related by a homeomorphism fixing that boundary. Take the identity on all unchanged tetrahedra and this boundary-fixing homeomorphism on each region. They agree on every identification between pieces, including identifications between two selected regions. They therefore descend to a homeomorphism of the quotient.

Replacing one filling leaves every other region's three tetrahedra and internal-face labels intact. Only its already specified external attachments are transferred. Its central edge still has exactly its three old occurrences: no boundary edge can become that central edge under a boundary-fixing replacement. Thus the remaining sites stay legal. The final quotient consists of the same pieces and the same attachment maps, independently of order. Ordering surviving input tetrahedra first and then each region's two outputs gives an explicit common indexing.

On a simply connected formal bipyramid, a closed integral edge cochain is the difference of five integral vertex heights, unique up to one additive constant. Coherent transport restricts these same heights to the new tetrahedra. Its values on all boundary edges are unchanged, so simultaneous transports agree on paired facets. Each region's height differences are unaffected by other replacements. Applying the local coordinate formula therefore gives the same final rows in every order. Finally, the local cell-count difference of a replacement is supported within its formal region and relative boundary; adding these differences, or telescoping through any valid order, proves additivity of the two raw changes. $\square$

4.2 The bounded conflict graph

Definition 4.2. The collapse conflict graph has one vertex for each distinct legal degree-three central *global edge*. Two vertices are adjacent when their regions share a tetrahedron. Candidates are deduplicated by the global central edge, not by its three local occurrences.

Theorem 4.3 (Nine-neighbour bound). *The collapse conflict graph has maximum degree at most nine. If M distinct legal candidates are currently available, every greedy maximal set of pairwise tetrahedron-disjoint regions has size at least $\lceil M/10 \rceil$.*

Proof. A formal three-tetrahedron bipyramid has ten edge types: its central edge AB , the three belt edges, and the six apex–belt edges. A candidate sharing one of its tetrahedra has its central edge represented by one of these ten local types. Distinct candidates with the same global edge have already been identified. The central type accounts for the original candidate, leaving at most nine others. Global identifications among boundary edges can only reduce this number. The old central edge cannot be identified with an additional boundary-edge occurrence, because its degree is exactly three and those three occurrences are already present.

In greedy maximal selection, each chosen vertex accounts for itself and at most nine unchosen neighbours. The closed neighbourhoods of the chosen vertices cover all M vertices, so $M \leq 10|E|$. $\square$

This is a statement about the current candidate set. New moves can appear after a batch, and old moves that conflict with a selected region can disappear. It does not identify the downward closure of the triangulation, nor does it prove that a particular greedy policy reaches an essential-disc witness.

4.3 Five-height raw scores

Write the belt heights in increasing order as $x \leq y \leq z$, and the apex heights as $L \leq U$. Define

$$g = \max(0, L - z) + \max(0, x - U), \quad (13)$$

$$D = \max(U, y) - \min(L, y) + \max(0, \min(U, x) - L) + \max(0, U - \max(L, z)). \quad (14)$$

The incoming transport theorem gives, in the 3–2 direction,

$$\Delta\chi_{\text{raw}} = 2g, \quad \Delta P_{\text{raw}} = -D, \quad g, D \geq 0. \quad (15)$$

For completeness, these formulas can be verified by writing the piece count in a tetrahedron as its height range. The old sum of the three ranges minus the new sum of two ranges is D . For Euler characteristic, the disappearing edge contributes $-|U - L|$ in the collapse difference; the three old internal-face ranges are replaced by the belt-face range, and the piece-count change completes the alternating cell sum. Sorting the three belt heights and clipping to $[L, U]$ gives $2g$.

By theorem 4.1, a batch S has raw gain $\sum_{i \in S} 2g_i$ and raw piece saving $\sum_{i \in S} D_i$. Each local computation uses a constant number of integer comparisons and additions, independently of the number of represented surface layers. The independent batch consumer does not rely on this gap formula: it recomputes the global normal cell counts at the two ends.

4.4 Encoding and validation cost

Let $k = |S|$. The combined output has $t - k$ tetrahedra. The certificate records k regions, each with three source tetrahedron indices and constant-size local labels. When transporting a cocycle it also records the five heights of each region and the final height and coordinate rows. There are no intermediate full triangulations.

Proposition 4.4 (Combined certificate size and construction work). *For input height-bit bound B , the combined geometric and cocycle certificate has $O((t + k)(B + \log(t + 2)))$ bits. Aside from finite-manifold validation and integer arithmetic cost, construction and relative-boundary replay use*

$O(t + k)$ indexed layout operations. The shipped producer performs three full manifold preparations for one batch: one for source discovery and two in its independent source/target consumer. This count is independent of k .

Proof. At most $3k$ old tetrahedra and $6k$ exposed old boundary facets need region records. Scan source tetrahedra once to assign new indices to survivors and to order regions by their least old index. Append two new tetrahedra per region. Every unchanged face and every region port is processed a bounded number of times. A face joining two regions is handled by composing its original permutation with the two old-to-new facet maps, so it does not require a search through other regions.

Each new tetrahedron uses four of the old region's five heights. Its edge differences are boundary-edge differences that already existed before collapse, so their bit length does not increase. The number and bit length of all stored rows give the stated encoding bound. The consumer reconstructs old and new ports once, validates the two finite manifolds once each, and compares their attachment signatures. The implementation's own source preparation adds the third call. $\square$

A literal sequential transcript can instead contain $\Theta(\sum_{j=1}^k (t - j)(B + \log t))$ bits because each step repeats the whole current triangulation and normal vector. On families with $k = \Theta(t)$ this is quadratic in t before counting bit lengths. This comparison concerns the existing full-state transcript interface; a separately designed persistent sequential transcript could also avoid that duplication. The combined certificate gives a direct implementation of the smaller interface now.

5 Dynamic vertex links and optimal peeled sub-batches

5.1 From a static prefix to repeated exact updates

The incoming geometric-transport report proves that thirteen corner records suffice to score one legal collapse after a shared preparation [4]. The bound is exact in its intended role: a move deletes twelve corner records and inserts eight. If an old record survives at an affected global vertex, the least survivor must occur among its thirteen least old records. Otherwise all thirteen would have been deleted. Equal coordinate values are ordered by distinct corner identities, so ties do not invalidate this pigeonhole argument.

Retaining only those thirteen records across repeated commits is incorrect. After deleting twelve records twice, the next true minimum can lie beyond every originally retained record. The new data structure keeps all live records in an ordered balanced tree and caches a short prefix at every node.

Theorem 5.1 (Dynamic corner-prefix maintenance). *Start with N local corner records, each carrying a stable identity, a global vertex label, and a nonnegative triangle coordinate. Consider a sequence of k independently justified local replacements, each deleting at most twelve old records and inserting at most eight new ones, preserving the global vertex classes and their link types. There is a deterministic data structure with the following bounds.*

- (i) *Construction uses $O(N \log(N + 2))$ comparisons and $O(N)$ stored records.*
- (ii) *Every hypothetical singleton replacement has its exact affected vertex minima and peeled score computed in $O(1)$ integer operations after its local record lists are known.*
- (iii) *Every committed replacement updates the data structure in $O(\log(N + k + 2))$ tree operations per touched record.*

- (iv) *Total maintenance, excluding geometric validation, candidate discovery, and literal emission of full vectors, uses $O((N + k) \log(N + k + 2))$ tree operations and $O(N + k)$ peak stored records.*

The operation count is independent of the numeric surface weight.

Proof. For each global vertex, order its records by the pair (triangle value, stable corner identity). Build an AVL tree. At each node retain its height, subtree size, and first thirteen keys. The prefix is obtained from the left child's prefix, the node key, and the right child's prefix, then truncated to length thirteen. Since thirteen is fixed, recomputing it costs constant work.

Insertion and deletion change only a root path and a bounded number of rotations per affected level. Ordinary AVL height bounds give $O(\log(N + k + 2))$ operations, and each rotation refreshes only constantly many short prefixes. A hypothetical singleton removes at most twelve identities; scanning the cached prefix gives the smallest old survivor. Compare that value with the at most eight inserted values. There are at most twenty affected global vertices in the abstract edit model, and at most five in an actual bipyramid. The exact minima, corner counts, and the two identities in (1) give its score.

All committed records are updated in the complete tree, so no future query relies on an exhausted old prefix. There are $O(N + k)$ total record insertions and deletions; summing the path bounds proves the maintenance estimate. Stable identities are never recycled. The canonical tetrahedron permutation in a batch certificate maps every surviving record, and fresh identities label newly inserted corners. $\square$

For a batch of k hypothetical moves, the union can delete $12k$ records, so a thirteen-record cache alone is insufficient. The implementation uses an in-order traversal at each affected vertex, stopping at the first record not in the deletion set. There are at most $12k$ deleted records to skip plus one survivor per affected vertex. This gives $O(k \log(N + k + 2))$ tree work for collective scoring, without modifying the live state. A committed batch performs the same ordinary tree updates as its component moves.

5.2 Local minima under coherent collapse

Fix one formal 3–2 region with gap g from (13). For a formal vertex, consider the minimum of its triangle coordinates over all corners belonging to this region. At a belt vertex, this regional minimum is unchanged. At either apex, it increases by exactly g . These are local identities, including coincident heights. One way to see the belt identity is to express a triangle coordinate at height h as the distance from h to the interval spanned by the other three heights. The relevant old and new intervals have the same union. For the apices, the formulas are explicit. With the belt heights $x \leq y \leq z$ and apex heights $L \leq U$, the old and new regional minima at the lower apex are

$$a_L = (\min(x, U) - L)_+, \quad b_L = (x - L)_+ + (L - z)_+,$$

and at the upper apex they are

$$a_U = (U - \max(z, L))_+, \quad b_U = (x - U)_+ + (U - z)_+.$$

Subtracting gives $b_L - a_L = b_U - a_U = (x - U)_+ + (L - z)_+ = g$. For the belt identity, the two old comparison intervals share the apex interval, while the two new intervals share the interval between the other belt heights; both unions are the hull of the same four other heights. Thus the distance description proves the identity also when intervals collapse or heights coincide.

Lemma 5.2 (Regional minima with global identifications). *For a global vertex v incident to region i , let a_{vi} and b_{vi} be its old and new regional minima, taking the minimum over all formal occurrences representing v . Then*

$$0 \leq b_{vi} - a_{vi} \leq g_i.$$

If v is not a global image of either apex of the region, then $b_{vi} = a_{vi}$.

Proof. Each formal regional minimum either stays fixed or increases by g_i . If finitely many numbers each increase by a value in $[0, g_i]$, their minimum also increases by a value in this interval. A vertex having only belt occurrences has no increasing formal term. $\square$

Taking the additional minimum with unchanged exterior corners proves that every global m_v is nondecreasing in a coherent downward epoch. It does not make the peeled Euler characteristic nondecreasing in general: an interior vertex has link Euler characteristic two, and the correction can exceed the raw gain. The sharper boundary-only case will follow from the selection theorem below.

5.3 The exact interaction formula

Fix a tetrahedron-disjoint ground set E of currently legal moves. For a vertex v , let o_v be its minimum at corners outside all regions in E , with value $+\infty$ when none exist. Put

$$\gamma_v = \min(o_v, \min_{i \in E} b_{vi}), \quad (16)$$

where absent incidences have value $+\infty$. Every global vertex has a corner after all replacements, so γ_v is finite. For a selected subset $S \subseteq E$, disjointness and lemma 5.2 give

$$m_v(S) = \min(o_v, \min_{i \notin S} a_{vi}, \min_{i \in S} b_{vi}) = \min(\gamma_v, \min_{i \notin S} a_{vi}). \quad (17)$$

The second equality holds because adding b_{vi} for an unselected region cannot lower a minimum that already contains $a_{vi} \leq b_{vi}$.

Theorem 5.3 (Supermodular minima and submodular peeled gain). *On any fixed tetrahedron-disjoint ground set E , $m_v(S)$ is monotone and supermodular for every global vertex. Consequently*

$$F(S) = \sum_{i \in S} 2g_i - \sum_v \lambda_v(m_v(S) - m_v(\emptyset)) \quad (18)$$

is submodular, and

$$F(S) \leq \sum_{i \in S} F(\{i\}).$$

The function F need not be monotone when interior vertices are present.

Proof. List, in increasing order, the lower endpoint, all distinct old regional minima between it and γ_v , and γ_v itself:

$$d_0 = m_v(\emptyset) < d_1 < \dots < d_q = \gamma_v.$$

When the endpoints coincide use the empty sum. Put $D_j = \{i \in E : a_{vi} < d_j\}$. Then the exact finite decomposition is

$$m_v(S) = d_0 + \sum_{j=1}^q (d_j - d_{j-1}) \mathbf{1}[D_j \subseteq S].$$

Indeed, the minimum reaches threshold d_j precisely when every old regional value below that threshold has been removed. The indicator of $D_j \subseteq S$ is supermodular: if two sets both contain D_j

then so do their intersection and union; if exactly one does, their union does; and if neither does the supermodular inequality is automatic. The displayed sum has nonnegative coefficients and includes the final jump to γ_v , so it proves supermodularity. Monotonicity is also immediate from (17).

The first term in (18) is modular and every λ_v is positive. Subtracting the supermodular correction gives submodularity. Since $F(\emptyset) = 0$, repeated application of its diminishing-returns inequality to singleton additions gives the last displayed bound. $\square$

5.4 A realized strict interaction

The retained fixture has 32 tetrahedra and one torus boundary component. Its construction starts with the barycentric subdivision of a one-tetrahedron solid torus. Attach an eight-tetrahedron ball along an embedded boundary triangle. The ball is the cone from a vertex v over the boundary of a tetrahedron after subdividing two of its faces with new vertices B_1, B_2 . The two degree-three edges vB_1, vB_2 have disjoint tetrahedron stars.

Give v height zero, B_1, B_2 height one, and all four original link vertices height minus one. Extend the potential constantly over the solid-torus part. These heights are an actual integral global vertex potential; the example uses the zero cohomology class and makes no claim to be a primitive meridian fibre. Its purpose is to test the universal scoring formula on valid ambient geometry.

State	Raw χ	Vertex-link correction	Peeled χ
Before either collapse	2	2	0
First collapse only	4	3	1
Second collapse only	4	3	1
Both collapses	6	6	0

Either single move leaves a zero triangle coordinate at v in the other region. Performing both removes all such corners, increasing m_v by one; its sphere link contributes the extra correction two. Thus the two singleton gains sum to two whereas the batch gain is zero. The package includes the complete face pairings, heights, construction, and native independent replays.

5.5 Exact selection by weighted matching

Submodularity alone does not provide an exact polynomial maximisation algorithm. Here a stronger property of coherent collapse does: each move can improve a regional minimum only at its two apices, and the reward at either global apex is bounded by that move's raw Euler gain. This reduces the entire interaction between disjoint regions to an ordinary weighted matching problem.

Continue with the fixed tetrahedron-disjoint family E and notation of (16)–(18). Put $k = |E|$, $r_i = 2g_i$, and define

$$c_{vi} = \lambda_v(\gamma_v - a_{vi})_+, \quad (\gamma_v - \infty)_+ := 0. \quad (19)$$

An omitted move preserves its old regional minimum; c_{vi} measures the resulting saving in the vertex-link correction, relative to performing all moves. Savings from several omitted moves at the same vertex combine by their maximum.

Lemma 5.4 (Two endpoints and a cost bound). *For each move i , at most two distinct global vertices have $c_{vi} > 0$, and every reward satisfies $0 \leq c_{vi} \leq r_i$. If v is a boundary vertex, the sharper bound $c_{vi} \leq g_i$ holds. For $R = E \setminus S$, the exact gain satisfies*

$$\begin{aligned} F(E \setminus R) &= F(E) + H(R), \\ H(R) &= \sum_v \max(\{c_{vi} : i \in R\} \cup \{0\}) - \sum_{i \in R} r_i. \end{aligned} \quad (20)$$

Proof. By definition $\gamma_v \leq b_{vi}$ at every incidence. Thus lemma 5.2 gives

$$c_{vi} \leq \lambda_v(b_{vi} - a_{vi}) \leq \lambda_v g_i \leq 2g_i = r_i.$$

A positive reward requires $b_{vi} > a_{vi}$, so v must be a global image of an apex. There are at most two such images, including when formal vertices are identified. At a boundary vertex $\lambda_v = 1$.

Equation (17) implies

$$\lambda_v(\gamma_v - m_v(E \setminus R)) = \max(\{c_{vi} : i \in R\} \cup \{0\}).$$

Subtracting the raw gains of omitted moves in (18) proves (20). $\square$

Form a weighted multigraph on the global vertices. A move i with two distinct positive-reward endpoints u, v supplies an edge of weight

$$w_i = c_{ui} + c_{vi} - r_i. \quad (21)$$

Discard edges of weight at most zero. Discard moves with fewer than two distinct positive-reward endpoints. In particular, two formal apices representing the same global vertex do not supply two collectable rewards: their image is one endpoint, and its reward occurs once in (20). Among parallel edges retain the largest weight; ties may use the least move index. Denote the resulting simple weighted graph by G_E .

Theorem 5.5 (Optimal peeled subbatch by matching). *Let E be a fixed tetrahedron-disjoint family of certified coherent 3–2 moves in a finite compact 3-manifold triangulation whose vertex links are spheres or discs. Then*

$$\max_{S \subseteq E} F(S) = F(E) + \max_{\text{matchings } M \subseteq G_E} \sum_{i \in M} w_i. \quad (22)$$

An optimal matching gives an optimal subbatch by omitting its moves and retaining every other move of E . Among equal optimum peeled scores, a maximum-weight matching of minimum cardinality gives a subbatch retaining the largest possible number of moves.

Consequently an exact optimum over all 2^k subbatches is computable in polynomial bit complexity. For integral weights the secondary cardinality rule is obtained by maximising

$$\tilde{w}_i = (k + 1)w_i - 1. \quad (23)$$

Proof. Choose a subset $R \subseteq E$ maximising $H(R)$ and, among these, having smallest cardinality. For a selected move $i \in R$, let U_i be the set of vertices at which i is the unique positive maximiser of the selected rewards. Removing i loses no reward outside U_i , and loses at most c_{vi} at $v \in U_i$. It also saves its cost r_i .

If $|U_i| \leq 1$, the total lost reward is at most r_i by lemma 5.4. Removing i would therefore not decrease H , contradicting the choice of R with smallest cardinality. Each move in R must consequently uniquely maximise the rewards at two distinct vertices. A move has at most two positive-reward endpoints, so these are precisely its endpoints. Two moves of R cannot share an endpoint: they cannot both be its unique maximiser. Therefore R is a matching and

$$H(R) = \sum_{i \in R} (c_{ui} + c_{vi} - r_i) = \sum_{i \in R} w_i.$$

Every one of its weights is strictly positive, since deleting a nonpositive edge cannot decrease H and reduces cardinality.

Conversely, for any matching the endpoint rewards are collected independently, and its value of H is exactly the sum of its edge weights. Replacing a parallel edge by one of greater weight never hurts feasibility or cardinality. Thus the preceding reductions to G_E preserve the optimum and a smallest-cardinality optimum. Equation (20) proves (22).

Classical maximum-weight matching is polynomial [6]. For the tie rule, a difference of one in integer total weight becomes a difference of $k + 1$ after scaling; any two matching cardinalities differ by at most k . Maximising the transformed total therefore first maximises the original total, then minimises the number of edges. This adds only $O(\log(k + 1))$ bits to the edge weights. $\square$

The cardinality rule needs the actual transformation (23). Deleting zero-weight edges alone is insufficient: two positive edges of weight one can tie a single edge of weight two. The smaller matching gives the same best Euler score while performing one more Pachner reduction.

The theorem concerns the fixed family E . It does not solve the additional packing problem of choosing an optimal disjoint family from all overlapping candidates. Also, attaining the largest peeled Euler value is a local policy objective; it does not authenticate an essential disc or prove that the resulting fibre has a desired connected component.

5.6 Interior vertices bound the number of omissions

Let p_E^+ be the number of interior global vertices incident to at least one positive reward c_{vi} . It is at most the total number p of interior global vertices of the triangulation.

Corollary 5.6 (At most one omission per relevant interior vertex). *There exists an optimal subbatch retaining at least*

$$|E| - p_E^+ \geq |E| - p$$

moves. The minimum-omission rule in theorem 5.5 satisfies this bound. With no interior vertices, executing all moves in E is optimal. With one relevant interior vertex, at most one move need be omitted.

Proof. If both endpoints of an edge are boundary vertices, then $c_{ui} + c_{vi} \leq g_i + g_i = r_i$, so its weight is not positive. Every edge of G_E therefore touches an interior vertex with positive reward. Edges of a matching have disjoint endpoint sets, so such a matching has at most p_E^+ edges. Apply theorem 5.5. $\square$

When all vertices are boundary vertices, there is also a sequential statement: every legal coherent 3–2 move has nonnegative peeled Euler gain. Its raw gain is $2g_i$, and the total increase of the vertex-link correction is at most $g_i + g_i$. This remains valid when the two apices represent one global vertex. Thus any sequence of such moves is nondecreasing in peeled Euler characteristic.

For M distinct eligible degree-three central edges, the degree-nine conflict bound supplies a greedy tetrahedron-disjoint family E with $|E| \geq \lceil M/10 \rceil$. Optimal selection within that family retains at least

$$\max\{0, \lceil M/10 \rceil - p\} \tag{24}$$

moves. Its total peeled gain is nonnegative, because the empty subbatch is among the choices. A bound on the number of retained moves and a bound on the score are distinct conclusions; neither requires summing singleton peeled gains.

Corollary 5.7 (The maintained two-pole pulling source). *Assume the validated one-component PD is converted to the canonical compact exterior by the maintained **pulling** subdivision, with its source-construction contract accepted. That triangulation has exactly two interior global vertices.*

This count remains two under the certified vertex-preserving 2–3 and 3–2 replacements considered here. For every fixed disjoint coherent 3–2 family E along such a trajectory, an optimal subbatch retains at least $|E| - 2$ moves. Starting with M candidates and the greedy packing therefore retains at least

$$\max\{0, \lceil M/10 \rceil - 2\}.$$

In particular, $M \geq 21$ guarantees at least one retained move at an optimum of the packed family’s peeled objective.

Proof. The source construction retains the north and south poles of the crossing-cell decomposition, each with a sphere link. Its pulling subdivision introduces only the six directed-edge cut points per cell, all on the truncation boundary. It introduces no face centres or cell centres. The gluing maps preserve the two pole labels and the cut-point locus; the source theorem identifies exactly two global poles and a single compact torus boundary [1]. Consequently these poles are the only interior vertices. Certified relative-boundary moves preserve the global vertex classes and their link types. Apply corollary 5.6 and (24). $\square$

This corollary is specific to the maintained pulling source and its preserved vertex classes. The legacy centred subdivision inserts additional interior vertices and does not have the same constant bound. Arbitrary simplification routines would also need their own proof of the vertex-count property.

5.7 Linear selection for the two-pole source

The constant omission bound has an algorithmic consequence stronger than applying a general blossom algorithm. Every positive edge of G_E meets an interior vertex. Thus the two poles form a vertex cover of the positive reward graph: every edge touches this set.

Theorem 5.8 (Matching with an at-most-two-vertex cover). *Suppose an explicit positive-weight graph is supplied together with a vertex cover C of size at most two. A maximum-weight matching of minimum cardinality can be found using a linear number of edge inspections, integer comparisons and arithmetic operations. For the two-pole pulling source, this gives an $O(k)$ matching-selection step once its reward graph has been materialized, with no external matching backend.*

Proof. First check that C has at most two distinct vertices and meets every edge. Apply the transformed weights (23). A matching has at most two edges, because each of its edges meets C and matching edges cannot share a cover vertex.

If $|C| \leq 1$, the best matching is empty or consists of a single largest-weight edge. Suppose $C = \{u, v\}$. A two-edge matching must consist of ux and vy , where $x, y \notin C$ and $x \neq y$. An edge uv can only occur in a singleton matching. During one scan retain the best singleton edge, the two best edges from u to distinct vertices outside C , and the analogous two edges from v . Parallel edges have already been compressed; equivalently one may maintain the best edge per retained outside endpoint during the scan. Keeping two entries at each pole requires constant work per edge.

For any admissible pair ux, vy , at most one of the two retained outside endpoints at u equals y . Hence either ux is already retained or it can be replaced by a retained edge of no smaller transformed weight whose outside endpoint differs from y . After this replacement, apply the same argument at v , holding the chosen u edge fixed. Thus an optimum two-edge matching is among the at most four combinations of the retained lists. Compare these valid combinations against the best singleton and the empty matching. This is constant further work after the scan. Maximising the transformed sum gives precisely the required primary and secondary objectives. $\square$

The supplied cover is an algorithmic precondition that the implementation checks against every positive edge. It is not accepted as an unchecked claim about the triangulation. The geometric wrapper obtains its candidate cover from the actual interior vertex classes and uses this path only when at most two relevant classes occur. Repeated endpoints, loops, parallel edges, zero gaps and equal-weight ties are handled by the preceding reward reduction and the same exact tie transform.

The linear bound counts operations on the already formed graph. It is not a linear bit bound independent of coordinate size, nor a bound for geometric preparation or collective minimum extraction. Each transformed weight has the original coefficient bit length plus $O(\log(k + 1))$ bits. The implementation’s graph assembly uses ordinary dictionaries; the theorem’s graph-scan bound does not assert a worst-case constant bound for arbitrary Python hash-table operations.

5.8 Algorithmic trust and independently checkable optimality

The implementation forms the reward graph using exact integers. With a validated cover of size at most two it uses [theorem 5.8](#), entirely in the standard library. If the graph has no positive edges, the empty omission set is optimal without an external matching backend. For other inputs the general solver uses the optional NetworkX maximum-weight-matching routine. Its documented bound is $O(q^3)$ arithmetic operations for q graph vertices, with integer-only computations when the weights are integers [7]. This is separate from the nearly linear corner-tree maintenance bound. Integer bit lengths, graph construction, and geometry validation must also be counted.

The returned omission set has its feasibility and attained objective recomputed directly from (20). This establishes its value; maximum-weight optimality of the general result still relies on the matching backend. The two-cover path instead uses the explicit constant-size candidate comparison just proved. The independent geometric replay certifies the selected moves and their exact cocycle transport, regardless of whether the policy found the best subset.

An optional, independent optimality checker accepts a classical Edmonds dual witness [6]. For each vertex take a nonnegative rational price y_v , and for finitely many odd vertex sets B with $|B| \geq 3$ take nonnegative rational prices z_B . Check every edge uv of G_E against

$$y_u + y_v + \sum_{B \ni \{u,v\}} z_B \geq w_{uv}. \quad (25)$$

Then check that the proposed matching weight equals

$$\sum_v y_v + \sum_B \frac{|B| - 1}{2} z_B. \quad (26)$$

These checks prove optimality by weak duality: a matching uses each vertex at most once and contains at most $(|B| - 1)/2$ edges inside an odd set B . Summing (25) over any matching bounds its weight by (26). Equality for the proposed matching proves it is maximum. The verifier needs no assumption that the listed odd sets are laminar and runs in time polynomial in the explicit certificate length.

The serialized certificate uses one positive integer denominator and nonnegative integer numerators, so these checks are exact. The current NetworkX path does not export such a witness; the checker is available for independently supplied certificates. It certifies the primary score optimum, not the secondary minimum-omission tie rule. Small instances are additionally cross-checked by an explicitly bounded exhaustive reference solver; this testing method is not the polynomial algorithm.

The resulting theorem removes an exponential enumeration of subbatches from this particular coherent policy problem. It does not bound the number of hierarchy stages, guarantee discovery of useful coherent regions, or establish the complete recognizer’s sought quasipolynomial running time.

6 Implementation, certificates, and integration

6.1 Additive modules and inherited dependencies

The implementation is an additive research overlay on the pinned revision. It does not replace the recognizer’s production decision policy or its complete fallback. This is a deliberate interface choice: the geometric results are ready to be used for supplied cocycles, whereas choosing a complete family of useful cocycles remains a separate theorem obligation. The active LP is useful for certificates and experiments but does not yet justify a performance-oriented default change.

Module	Responsibility
<code>normal_active.py</code>	Complete active-support certificates, conflict branching, and the finite normal-anchor wrapper.
<code>normal_active_verify.py</code>	Solver-free arithmetic, branching, anchor, and source checks.
<code>pachner_batch.py</code>	Legal-region census, disjoint packing, and one simultaneous geometric/cocycle producer.
<code>pachner_batch_verify.py</code>	Independent port, manifold, cocycle, and normal-coordinate replay.
<code>corner_minima.py</code>	Deterministic AVL trees with thirteen-record summaries, exact hypothetical edits, and committed updates.
<code>cocycle_peeling.py</code>	Stable geometric identities, complete candidate scoring, collective scoring, and optimum packed subbatches.
<code>cocycle_peeling_verify.py</code>	Full independent endpoint reconstruction of each batch’s six raw and peeled score quantities.
<code>peeled_batch_selection.py</code>	Exact reward reduction, the small-cover selector, optional general matching, and certificate-value/dual checks.

Three necessary modules are inherited without claiming new authorship of their algorithms: `exact_lp.py` from the incoming dual report, and `cocycle_transport.py` with its verifier from the incoming transport report. They are absent from the pinned maintained tree and are included explicitly in the overlay. All other supporting maintained files retain their pinned contents. The integration manifest distinguishes these dependencies, new code, tests, research scripts, and supporting fixtures.

6.2 State ownership and cancellation

`CocyclePeelingState` owns a private triangulation, coherent heights, stable global-vertex names, and stable corner identifiers. Canonical tetrahedron renumbering changes positions but preserves the identities of surviving records. Each inserted corner obtains a fresh identifier. The ordered key is the pair consisting of its triangle multiplicity and this identifier; equal multiplicities therefore cause no ambiguity in deletion or prefix maintenance.

Candidate objects are scoped to the current immutable state. The cheap collective scorer assumes an unmodified subset of the current legal census; it is not an independent validator of user-edited region dictionaries. Committing a batch reconstructs and verifies the actual geometric moves before using their score to change the cache. The diagnostic `audit_index()` also rebuilds the global vertex equivalence relation and boundary/interior link types independently, then checks all records and AVL invariants. This protects against the subtler possibility of internally consistent records assigned to incorrect stable vertex classes.

Work callbacks propagate cancellation. If interruption occurs during a cache commit, the object is explicitly invalidated; a subsequent query requires reconstruction instead of using partially updated summaries. Neither callback limits nor LP pivot limits are interpreted as mathematical negative answers. Integer serialization uses the maintained codec, including its large hexadecimal representation, so a 20,001-bit scale is not limited by Python’s decimal-string conversion guard.

6.3 Three different kinds of verification

Arithmetic and search coverage. The active-support consumer recomputes all residuals with exact rational arithmetic. The conflict-tree consumer checks the selected disjunction and both children at a negative branch. The normal wrapper reconstructs the anchor product from the source; a proof cannot omit an inconvenient anchor. It returns a positive-Euler normal vector or a certified absence for that specific anchored search problem. It does not certify that a positive vector is connected.

Geometry and attained score. The batch consumer imports no batch producer. It reconstructs the two manifolds, all local region labels, and both ends of every boundary-port attachment. It checks the transported cochain and normal coordinates, and compares global cell counts. The separate peeling consumer imports neither the dynamic index nor the peeling producer: it scans both endpoints and checks raw Euler gain, raw piece saving, link Euler change, link piece change, peeled Euler gain, and peeled piece saving. This deliberately more expensive endpoint verification is the small trusted boundary around the fast incremental calculations.

Policy optimality and the topological endpoint. The matching reduction is an exact mathematical algorithm. The small-cover path enumerates the constant number of possible matching forms after a linear summary pass. The general path relies on the matching backend unless an independent dual certificate is supplied. Recomputing an attained score alone does not certify maximality. In either case, a suboptimal policy choice cannot make an invalid geometric transition pass its consumer.

After obtaining a promising supplied normal vector, the maintained `normal_compressing_disc_count` remains the relevant component query. Its independently replayable certificate peels vertex links and uses the existing compressed component machinery. The positive control in the active-search audit passes through this endpoint. Neither a favorable peeled Euler score nor the matching output is substituted for a compressing disc with essential boundary. Finally, turning that disc into an unknot verdict requires the accepted diagram-to-exterior correspondence.

6.4 A bounded, replayable descent driver

The research driver composes the complete census, greedy packing, exact subbatch selection, and verified commit until it stalls or reaches a specified round limit. It requires at most two interior vertices and keeps the original triangulation and cocycle bound to its certificate. A separate chain checker replays each endpoint through the geometry and peeling consumers. A stopping status is a producer diagnostic, not a certificate that all useful global geometric paths have been exhausted. Optional final disc counting uses the maintained component certificate on the actual final coherent vector. Work-limit and callback exceptions propagate without returning a partially accepted trajectory. This driver makes the collapse-only algorithm of [proposition 8.1](#) concrete without asserting a new complete recognition policy.

6.5 Recommended order of integration

First integrate the independent consumers and retained fixtures, then the simultaneous batch producer. The fixed-site equivalence tests isolate this change from policy questions. Next integrate the minima index and compare its entire candidate census with full singleton replay. Enable optimum subbatch selection only after checking the actual global vertex types; the two-interior-vertex fast path must not be inferred from a filename or knot label. Keep the active LP optional until a sparse or reused-basis producer beats the simpler probes on matched inputs.

The supplied patch only adds files. The supporting snapshot makes the experiments reproducible without assuming that the three incoming packages have already been integrated. If a later repository revision has introduced files under the same names, reviewers should reconcile those additions against their recorded dependency lineage instead of overwriting them.

7 Reproducible experiments and adverse results

7.1 Environment and measurement design

All retained measurements use Python 3.12.14 on Linux x86_64. The geometry, active LP, small-cover selector, and independent replay paths use Python’s standard library and the included maintained modules. General weighted matching was checked with NetworkX 3.7. Regina 7.4.1 was used to generate the original barycentric-torus seed for the counterexample; its full finite face pairing is retained, so replay and reconstruction from that seed do not require Regina. Matplotlib is required only to regenerate the figure. The environment manifest records the versions actually available.

These are single-machine subroutine experiments, with no distributional claim about arbitrary knots. Batch and complete-census timings use three paired rounds, alternating which arm runs first. Tables report the median of each arm; speedup is the ratio of those medians. Every raw sample and callback count is retained. The active-support audit uses one timing per request and reports that different design explicitly. The small timings are not estimates with a justified confidence interval.

7.2 Same moves, simultaneous versus sequential certificates

The fixed-batch benchmark has four source families and six batch sizes $k \in \{1, 4, 8, 16, 32, 64\}$, giving 24 cases. The families are a layered solid torus and diagram-bound unknot, trefoil, and figure-eight exteriors. Source construction and explicit 2–3 inflations supply the same disjoint collapse sites to both arms. Inflations make the workload controlled; their setup is not included in either timed arm. Larger diagram-bound cases use extra cancelling braid letters to provide enough initial cells. Consequently the table describes fixed local workloads, not a naturally observed distribution of reduction batches.

The sequential arm composes the established checked individual Pachner move and coherent transport, retaining full intermediate states. The batch arm constructs one final state and one combined witness. Producer timing includes each arm’s own independent replay. Standalone verifier timing is measured separately. In all cases the final face pairings and height rows agree exactly after the explicit tetrahedron permutation. There are 144 timed producer runs and 144 timed standalone verifier runs, all successful.

The corresponding literal JSON witness sizes, sequential versus batch, are 2,833,339 versus 47,725 bytes for the layered torus; 2,104,416 versus 34,981 for the unknot; 2,104,680 versus 35,003 for the trefoil; and 2,346,159 versus 38,835 for the figure-eight. The ratios are 59.37, 60.16, 60.13, and 60.41. They measure the actual retained interfaces, not an information-theoretic lower bound

Family, $k = 64$	Input t	Sequential (s)	Batch (s)	Producer ratio	Replay ratio
Layered torus	192	3.3626	0.02740	122.71	89.13
Unknot	204	3.2253	0.02711	118.96	95.89
Trefoil	204	3.3530	0.02855	117.42	96.26
Figure-eight	224	3.7267	0.03090	120.61	88.55

Table 2: Paired median times for the same 64 collapses. A ratio above one favours simultaneous processing. Producer and replay ratios refer to separate timing columns in the retained JSON.

on every possible sequential encoding. The $k = 1$ cases are also retained: their producer ratios range from 1.82 to 1.90, replay ratios from 1.41 to 1.57, and byte ratios from 1.02 to 1.20. Eliminating redundant preparations helps even at one move. There is no timing regression in this particular fixed-batch workload.

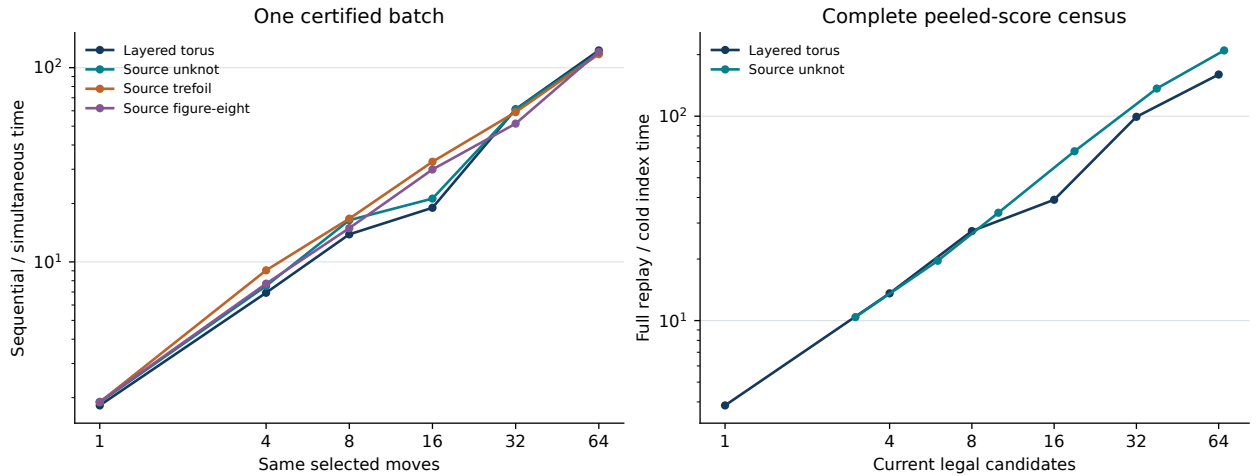


Figure 1: Observed subroutine ratios, with three alternating paired rounds per point. Left: simultaneous checked producer versus the existing literal sequential composition on the same selected sites. Right: a freshly built minima index and complete candidate census versus full individual geometric replay of every candidate. Lines connect measured points; they are not fitted asymptotic bounds.

7.3 A cold index plus the entire peeled-score census

This comparison starts the dynamic state from scratch on every timed run. It includes the index construction and complete current candidate census. The reference independently prepares the same input, performs every candidate with the individual geometric producer and transport consumer, and recomputes all vertex-link minima. The entire ordered list of site, peeled Euler gain, and peeled piece saving agrees exactly. It is not a comparison against a previously optimised static thirteen-prefix implementation; the incoming report supplied a theorem for that pass, not such an implementation.

The tree-maintenance theorem concerns repeated local edits. These timings test a more conservative cold-start query; they do not isolate an amortised update speedup. A separate fresh-geometry audit checks 23 committed batches and 45 moves across four families, including stable-vertex/fresh-root bijections, link types, every triangle record, all thirteen-key prefixes, and exact collective score

Family	Inserted sites	t	Census M	Full replay (s)	Cold index (s)	Ratio
Layered	1	3	1	0.00135	0.00035	3.85
Layered	4	12	4	0.01738	0.00128	13.59
Layered	8	24	8	0.06480	0.00236	27.42
Layered	16	48	16	0.30292	0.00777	39.00
Layered	32	96	32	1.14917	0.01157	99.30
Layered	64	192	64	4.74531	0.02968	159.86
Unknot	1	21	3	0.02030	0.00195	10.40
Unknot	4	24	6	0.04827	0.00246	19.62
Unknot	8	28	10	0.09583	0.00284	33.69
Unknot	16	76	19	0.50188	0.00745	67.37
Unknot	32	132	38	1.71692	0.01258	136.53
Unknot	64	204	67	5.09894	0.02432	209.69

Table 3: Complete peeled candidate scoring, including a fresh index on the faster arm. The census can include legal sites beyond the inserted ones.

agreement. All checks pass.

7.4 The active-support lift reduces calls but loses elapsed time

The active audit contains 20 partial-mask cases across seven genuine finite triangulations. Seventeen have positive objective and compare complete support discovery by efficient unary probes with the homogeneous lifted query. The unary method skips coordinates whose activity is already witnessed by an earlier LP point. This is therefore a meaningful small-LP baseline, not an artificially padded fixed number of queries.

Family	Masks	Unary calls	Lift calls	Unary (s)	Lift (s)	Lift/unary
Fibonacci, $t = 1$	2	6	2	0.00177	0.01644	9.30
Fibonacci, $t = 2$	3	16	3	0.02559	0.21252	8.30
Fibonacci, $t = 3$	3	24	3	0.17221	1.22396	7.11
Fibonacci, $t = 4$	3	27	3	0.42699	4.12959	9.67
Branching torus, $t = 5$	3	31	3	0.68977	16.10012	23.34
Boundary attachment, $t = 3$	3	26	3	0.10197	1.14782	11.26
Finite trefoil, $t = 5$	3	3	0	0.22226	—	—

Table 4: Time sums across the specified masks. The last row terminates after negative ordinary prechecks, so it does not run the lifted query.

Across the 17 positive paired cases, the unary method uses 130 LP calls, 1,418 pivots, and 1.418307 seconds; the lift uses 17 calls, 1,449 pivots, and 22.830456 seconds. The aggregate slowdown is 16.10-fold. The median individual slowdown is 8.84-fold, the range is 5.89–38.48-fold, and the lift wins no timing comparison. Similar pivot counts do not imply similar work because the lifted tableau is much larger. These data support the certificate interface but oppose enabling this dense implementation as a general speed optimisation.

A separate exploratory direct lift of the unrestricted five-tetrahedron trefoil root took 78.577357 seconds and 2,664 pivots to return nonpositive. It is a single exploratory observation, not a repeated paired measurement. The retained default audit uses a 25-second allowance per request and has no cap outcomes. The ordinary precheck prevents this unnecessary lifted query on the negative control.

7.5 Complete search controls and independent replay

Source	Precheck	Time (s)	Nodes	Branches	LP calls	Lifts
Fibonacci 1	yes	0.00105	3	0	3	0
Fibonacci 1	no	0.04512	5	2	5	5
Fibonacci 2	yes	0.17846	5	2	7	2
Fibonacci 2	no	0.48030	6	3	6	6
Fibonacci 3	yes	1.22946	6	3	9	3
Fibonacci 3	no	3.29603	7	4	7	7
Finite trefoil	yes	2.45246	20	0	20	0

Table 5: Seven full anchored searches. The Fibonacci controls return positive Euler; the finite trefoil proves no positive-Euler vector across its 20 anchors. Every result has an independently replayed certificate.

The replay script checks 37 support certificates and all seven complete normal-search certificates without importing an LP solver. Positive meridian controls are additionally checked through the maintained compressing-disc endpoint. The experimental active search does not include the earlier mandatory-type propagation, so its branching on Fibonacci inputs is not a measurement of the maintained recognizer’s policy. When no active query ran, an empty observed-rank statistic is not a proof that the root rank is zero.

The final maintained-plus-overlay test run reports 1368 tests passing. It includes 56 focused tests for the new interfaces. Tests cover all 24 orders of four disjoint sites, all six orders of three sites on each of three source exteriors, shared boundary facets, true overlapping-star rejection, malformed/corrupted proofs, producer-disabled consumer replay, exact large integer scales, callback cancellation, and input immutability. Randomised algebraic audits compare 600 bounded matching instances with exhaustive search, including 300 small-cover cases with the general backend disabled. Three initial full-run errors were missing historical fixture files; two byte-verified pinned fixtures were restored, the affected tests passed, and the complete suite was rerun. The retained logs distinguish that setup failure from the final verified result.

8 What these bounds do, and do not, imply asymptotically

8.1 A complete collapse-only trajectory is a polynomial subroutine

Let $V(t, B)$ bound the cost of the maintained finite-manifold and cocycle preparations on t tetrahedra with B -bit edge differences. Initial height rows are assumed gauged, for example by subtracting their first entry. If arbitrarily large common offsets are supplied, reading and normalising those integers is charged separately in their actual input bit length. We keep the preparation cost explicit. Let $M(B)$ stand for a safe bound on the relevant exact integer arithmetic cost; additions and comparisons are cheaper than general multiplication. A complete current census has $O(t)$ candidates, because there are at most $6t$ local edge occurrences. Sorting for greedy packing costs $O(t \log(t + 2))$ comparisons. The minima index and source-specific matching add polynomial work independent of expanded surface weight.

Proposition 8.1 (Cost and termination of the supplied local policy). *Start with a coherent cocycle on a valid triangulation with at most two interior vertices. At each round, form the complete legal 3–2 census, choose a greedy maximal disjoint family, select its optimal peeled-Euler subbatch with the minimum-omission rule, and commit it if nonempty. Stop otherwise. Every committed round is*

nondecreasing in peeled Euler characteristic; there are at most t_0 rounds and at most t_0 individual collapses, where t_0 is the initial tetrahedron count. The maximum encoded edge-difference bit length does not increase.

With polynomial V and exact arithmetic, this is a polynomial subroutine in (t_0, B_0) . A deliberately loose explicit bound is

$$O\left(t_0 V(t_0, B_0) + t_0^2 \log(t_0 + 2) M(B_0 + \log(t_0 + 2))\right), \quad (27)$$

assuming a nondecreasing upper bound V . The lifetime maintenance work of the corner index alone is only $O(t_0 \log(t_0 + 2))$ record operations.

Proof. The empty subbatch is available and has gain zero, so an optimum cannot have negative gain. Every committed move reduces the number of tetrahedra by one. Thus at most t_0 moves can be committed, and each nonempty round contains at least one. The commuting theorem preserves the vertex types, so the small-cover selection remains applicable. Every edge after a 3–2 move already occurred on the bipyramid boundary; its cocycle value is unchanged. This proves the bit statement.

Charge each round its complete census, packing, reward computation, linear small-cover matching, final emission, and constant number of preparations. Bounding every current size by t_0 and the number of rounds by t_0 gives (27). There are initially $4t_0$ corner records; each collapse deletes twelve and inserts eight, so the total number of index edits is $O(t_0)$. The balanced-tree theorem supplies the last assertion. $\square$

This proposition gives an honest finite local algorithm, including its stopping condition. It makes no claim that the terminal coherent vector contains a disc or that a stalled triangulation cannot be the unknot. Repeating an uncontrolled 2–3 repair phase is outside its proof. Likewise, recording every endpoint of every round can still cost quadratic space in t_0 ; the per-batch encoding theorem does not automatically supply a linear-size transcript for an entire trajectory.

8.2 A density hypothesis would give logarithmically many rounds

Corollary 8.2 (Conditional dense-census progress). *Suppose a collapse-only trajectory as above has at least αt eligible distinct central edges whenever its current size is $t \geq 40/\alpha$, where $0 < \alpha \leq 6$ is fixed. Then it reaches size below $40/\alpha$ in $O(\alpha^{-1} \log(t_0 + 2))$ committed rounds.*

Proof. The packed optimum retains at least

$$\lceil \alpha t / 10 \rceil - 2 \geq \alpha t / 20 \quad (t \geq 40/\alpha).$$

Thus the next size is at most $(1 - \alpha/20)t$. Repeated multiplication proves the bound, using $-\log(1 - \alpha/20) \geq \alpha/20$. $\square$

The density assumption is not established for all diagram exteriors or after arbitrary moves. A legal census can be sparse or empty. The corollary is useful because it specifies an exact geometric statement that would turn the batch-retention theorem into a round bound, without conflating raw score, peeled score, and the number of removed tetrahedra.

8.3 The normal-search ledger

The active-face theorem isolates the factor $\mathcal{A}(T)2^{\rho_{\max}}$. It supplies no universal reduction of this factor. In fact, the unrestricted source lower bound rules out the proposal that $\rho_{\max}$ is automatically polylogarithmic in crossing number. The anchor count also needs separate attention. In a one-vertex

finite triangulation it is $4t$; in a triangulation with many vertices it is the product of their corner degrees. The fact that there are only two *interior* vertices in the pulling source does not make that anchor product small: the same source has $4n'$ boundary vertices. Confusing the two vertex counts would erase a genuine source of combinatorial cost.

Complete-support certification, support minimisation of one admissible point, and proving a small active ambiguity rank are three different tasks. The first now has a compact interface. The second already has classical LP-based solutions [8]. The third fails at the unrestricted roots proved above. A promising parameter must therefore be defined after stronger certified restrictions, or the algorithm must be proved to reach an endpoint before traversing the large rank.

8.4 A sufficient quasipolynomial contract

For clarity, consider a future complete architecture with at most $R(n)$ outer stages. At each stage suppose it covers all required geometric choices by a tree of depth $h(n)$, branching at most $(ct(n)+c)^d$ ways per node for fixed positive constants c, d , and each node performs at most $A(n)$ normal queries with active rank at most $\rho(n)$. Suppose all state sizes are bounded by $t(n)$ and all arithmetic inputs by $B(n)$ bits, with polynomial local producers and consumers. A sufficient upper bound is

$$R(n)A(n)2^{\rho(n)}(ct(n)+c)^{O(h(n))}\text{poly}(t(n), B(n)). \quad (28)$$

For example, if t, B, R are polynomial in n , $\log A(n) = O((\log n)^a)$, $\rho(n) = O((\log n)^b)$, and $h(n) = O((\log n)^c)$ for fixed exponents, then (28) is

$$\exp\left(O((\log n)^{\max\{1, a, b, c+1\}})\right).$$

This is an elementary composition bound, not a theorem that those hypotheses hold. It displays the obligations a genuine quasipolynomial proof must discharge.

The present results improve the local polynomial factor and eliminate $2^{|E|}$ subbatch enumeration for a fixed disjoint family. They do not prove geometric witness coverage, the depth bound h , the restricted-rank bound ρ , or the anchor bound A . The explicit root obstruction also shows why replacing the last two obligations by an empirical observation would not suffice. Classical compressed component algorithms avoid dependence linear in expanded normal weight [11]; they do not bound the number of global choices by themselves.

9 Further research: statements worth proving next

The following questions are ordered by how directly a positive answer would improve the current algorithm. Each separates a mathematical target from an implementation experiment. Several may have negative answers; explicit counterfamilies would still prevent investment in an invalid complexity strategy.

9.1 Geometric progress and the correct search space

1. When must a useful collapse exist? Characterise coherent primitive-meridian fibres for which a legal 3–2 move or nonempty optimum packed subbatch exists before an essential disc is exposed. A strong target is a density or amortised density theorem of the form in [corollary 8.2](#). A weaker useful target is a bounded repair lemma: a stalled state admits at most $f(t)$ certified expansions followed by a batch that reduces a well-founded, explicitly encoded potential. Experiments should record empty censuses, small censuses, negative singleton scores, and stalls separately, because the matching theorem resolves only collective selection once candidates exist.

2. Can the two-pole structure survive more powerful reductions? The current fast selector survives vertex-preserving 2–3 and 3–2 moves. Crushing, collapsing boundary cells, idealisation, and handle normalisation can alter the vertex links or the representation. Prove an interface theorem retaining a bounded set of interior reward vertices, or identify an equally small parameter in the new representation. A finite two-pole triangulation and an ideal triangulation have different local link contracts; transferring formulas between them requires a proof.

3. Optimise the disjoint ground family, not only its subbatches. The degree-nine argument gives a constant-factor cardinality packing, and the new matching theorem exactly solves selection inside that packing. It leaves the choice among overlapping regions unresolved. Determine whether conflict graphs arising from legal bipyramids have additional structure beyond bounded degree that permits an exact or suitably certified approximation algorithm for the final peeled objective. An arbitrary bounded-degree graph does not acquire easy maximum-weight independent-set optimisation merely from its degree bound. Small geometric instances should compare every packing, every subbatch, and greedy packing followed by matching to identify the actual lost opportunity.

4. Is peeled Euler characteristic the right policy potential? The shared-vertex counterexample proves a failure of additivity on a genuine solid torus, but its supplied cocycle is exact. Investigate the same interaction under primitive meridional cohomology and boundary constraints. More broadly, seek a potential that combines tetrahedron count, peeled Euler characteristic, essential-boundary information, and compressed component structure. Its monotonicity must survive globally identified vertices and sphere components. A scalar score that increases while losing the relevant disc component would not establish witness coverage.

5. Bound the number of genuinely different geometric choices. Commutation identifies all orders of one fixed set of disjoint moves. It does not identify different sets or repairs through overlapping sites. Develop a canonical transition certificate, a confluence theorem on a restricted class, or a partial-order reduction that preserves at least one successful witness path. The target is a bound on distinct relevant states, with a proof that the quotient preserves the topological endpoint, rather than a reduction in the number of syntactic transcripts alone.

9.2 Admissibility, support, and certificate-efficient LP

6. Remove the explicit root obstruction only with a sound quotient. The local directions w_a change all three quadrilaterals together and are invisible to face-arc matching. Can one quotient or eliminate those directions while preserving admissible positive-Euler existence, including the triangle anchors and Euler objective? Their compatibility effect is nonlinear, so simply deleting them from a rank computation is unsound. A useful answer would give a reconstruction theorem from a quotient solution and a polynomial certificate of every excluded alternative. Alternatively, prove that a natural strengthened relaxation still has a linear ambiguity family; that would rule out another tempting route.

7. Relate active rank to mandatory propagation after restrictions. The incoming Fibonacci result has zero mandatory guess budget, while the unrestricted active rank here is linear. These parameters measure different things. Define a parameter for a *certified adaptive trajectory* after all currently forced types have been propagated, and ask whether it is polylogarithmic on a topologically meaningful class. The quantifiers matter: the existence of one easy witness is weaker

than a complete negative decision procedure. Any bound must include the dual certificates and incompatible branches needed to prove absence.

8. Produce the active-support certificate more cheaply. The current homogeneous lift adds $O(N + m)$ auxiliary structure to an already degenerate LP and is slower in the retained audit. Investigate sparse relative-interior methods, a reused primal/dual basis, incremental forced zero aggregation, or a hybrid that invokes the full lift only after a measured trigger. Basis reuse itself has established normal-surface precedents [10]; novelty should be measured by the resulting certificate size or verified work reduction. Benchmark rational bit growth, pivots, matrix density, calls, producer time, and replay time together.

9. Compress anchor coverage. Find a polynomial or quasipolynomial certificate representing the complete union of vertex-anchor cones without listing every anchor tuple. Shared subproofs, symmetry, vertex-link quotient coordinates, or a proof-producing elimination rule may help. A checker must reconstruct why every canonical normal vector is covered; a cache containing only visited anchors is not a negative certificate. The two-pole source is a useful test because it has few interior vertices but linearly many boundary vertices.

9.3 Compressed topology and a global complexity theorem

10. Make hierarchy transitions preserve all required markings. Relate the repository's compressed normal components and boundary moments to the boundary patterns and parallelity data needed by hierarchy methods. Specify what a transition certificate records, how it is independently verified, and how its encoded size changes. The 2026 hierarchy work offers concrete topological progress measures [13], but importing one of them requires a proved correspondence with this triangulation and cocycle implementation. Boundary curve normalisation must count every parameter required by its algorithm; the simplification parameter in the current fast-curve theorem cannot be suppressed [14].

11. Prove a bit-size invariant through repairs and regauging. Collapse-only transport has a nonincreasing edge-difference bound. Global regauging, LP extraction, expansions, and hierarchy repairs need a joint bound on integer and rational sizes. Establish a potential measuring encoding length as well as topological complexity. An algorithm may take few combinatorial steps but still create excessively large coordinates; counting arithmetic operations alone would miss that growth.

12. Export matching and transition optimality certificates. The general matching backend currently supplies a checked attained value without exporting an Edmonds dual. Add a producer for a sparse dual, including the integer scaling for minimum omissions, and verify it through the existing independent interface. For the two-cover case, a small certificate of the per-pole best endpoint choices should suffice. This is primarily a trust-boundary improvement; move correctness already has an independent geometric consumer.

13. Test whole-policy effects on source-bound hard families. The retained timing work fixes the same moves or the same candidate queries on both sides. The next experiment should compare complete policies with identical source diagrams, endpoint checks, and resource accounting. Include hard unknots, nontrivial knots escaping early filters, repeated expansions, and low-density stalls. Report failed budgets and regressions, not only successful reductions. Separate time to find a

witness, certificate size, and independent replay time. No finite corpus alone supplies a worst-case theorem, but it can identify which missing lemma would have practical value.

14. Close the cost ledger with explicit quantifiers. The eventual target is a theorem covering every input diagram and every state required by a complete search, with bounded transition depth, branching, anchors, and bit sizes as in (28). A resource-limited research module that returns an inconclusive status is useful engineering; it is not a total quasipolynomial decision algorithm. The most productive next theorem would bound a presently uncontrolled factor on a nontrivial, source-defined class and provide the certificate that makes that bound usable by the implementation.

A Package map and reproduction

The archive is rooted at `unknot_active_faces_batched_descent_20261009/`. The top-level README gives exact commands and the integration contract. The manifest records source revisions, incoming dependency hashes, and the identities of retained fixtures. Relative paths are used throughout the executable research scripts.

Path	Contents
<code>article/</code>	Complete modular L ^A T _E X source, final PDF, and vector performance figure.
<code>code/fast/</code>	Pinned supporting implementation, additive modules, focused and maintained tests, benchmark and replay scripts.
<code>code/synthesis/data/</code>	The two pinned historical fixtures needed by maintained test imports.
<code>code/reports/</code>	Historical oracle helpers required by existing tests, with provenance recorded.
<code>integration/</code>	Additive patch against the pinned tree, file lineage, and the patch-check result.
<code>evidence/geometry/</code>	All 24 fixed-batch inputs, source bindings, expansion records, both witnesses, and raw paired timing samples.
<code>evidence/counterexample/</code>	The realized shared-vertex example, portable construction, retained 24-tetrahedron seed, and batch witness.
<code>evidence/validation/</code>	Complete test logs, independent dynamic audit, cold-census timings, environment, and replay summaries.
<code>tools/</code>	Figure regeneration and package verification helpers.

From `code/fast/`, the fast certificate checks do not need an LP solver or optional graph library:

```
python -S active_research/replay.py
python -S batched_descent_research/replay_evidence.py
```

The full test suite is run with:

```
python -m unittest discover -s tests -v
```

NetworkX enables the five general matching tests; the source-specific small-cover tests still run with site packages disabled. Exact geometric and arithmetic replay use only included modules. The optional regeneration of the barycentric torus from its original triangulation library is separate from replay of the retained seed.

The checked producer benchmark can be rerun with the supplied `batched_descent_research/benchmark_batch.py`; it reconstructs source-bound inputs and performs alternating paired runs. Its command-line help describes size and output controls. The peeled-census benchmark takes the retained evidence directory as an explicit argument, and `active_research/benchmark.py` reconstructs its masks and support queries from the supplied source file. Reproducing exact elapsed times is not expected across hosts; reproducing exact face pairings, heights, supports, objective values, and valid certificates is the relevant correctness condition.

Compile the article from its directory with:

```
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The final PDF was rendered and visually checked in addition to successful L^AT_EX compilation. The package does not require downloading the three large incoming archives: the few unchanged prerequisite modules are included and explicitly attributed. It also does not include a new claim that the complete recognizer has quasipolynomial worst-case complexity.

B The general two-endpoint algebraic extension

The exported algebraic selector allows nonnegative costs r_i and at most two nonnegative endpoint rewards c_{vi} even when the geometric bound $c_{vi} \leq r_i$ is absent. This extension is not needed for the main geometric theorem. Here is the correctness argument for its implementation. Let

$$b_v = \max(0, \max_i(c_{vi} - r_i)).$$

A two-endpoint item i on distinct vertices u, v receives adjusted edge weight $c_{ui} + c_{vi} - r_i - b_u - b_v$. Discard nonpositive edges and retain the largest parallel edge. Then the optimum of H is the sum of the b_v plus the maximum adjusted matching weight.

To prove the upper bound, take any omitted set and assign each vertex's positive maximum reward to one attaining item. Delete items assigned no vertex; their costs are nonnegative. Each remaining item is assigned one or two vertices. A singly assigned item contributes at most b_v . The doubly assigned items form a matching on their assigned pairs and contribute their adjusted weights plus those vertices' baselines. Unused nonnegative baselines can only enlarge the bound. Dropping nonpositive adjusted edges can also only enlarge it.

For the reverse bound, choose a maximum adjusted matching and, for each unmatched vertex with positive baseline, choose a baseline-attaining item. Take the union of these items with the matching items. At each vertex the actual maximum reward is at least the assigned one; an item chosen more than once is paid for only once, so its actual total cost is at most the sum charged in the assignment. Thus H is at least the proposed matching-plus-baseline value. The upper bound proves equality. In the geometric case all baselines vanish, and this reduces to the main theorem. Outside that case the secondary matching-cardinality rule need not minimise the number of reconstructed singleton items; the API records this limit.

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